

Akash Kesav Shanmugaraj

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Available May 2026 – Dec 2026 for internship (open to extension)

SUMMARY

Machine-learning-focused Smart Manufacturing undergraduate at IIITDM Jabalpur (CGPA 7.7/10) with hands-on experience building and evaluating ML systems. Ran **12+ model experiments**, achieving up to **98% accuracy** on NLP classification and **83% accuracy** on semiconductor band-gap prediction. Built LangChain/LangGraph agent systems, shipped a MatterGen package on PyPI, and contributed to an accepted abstract at **Materials Today Conference (Elsevier, 2025)**.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Bachelor of Technology in Smart Manufacturing | CGPA: **7.7 / 10**

Expected May 2027
Jabalpur, Madhya Pradesh

EXPERIENCE

Rakumura IT Solutions

Jun 2025 – Nov 2025

AI/ML Backend Development Intern (Remote)

Chennai, Tamil Nadu

- Architected **multi-agent AI systems** using **2 orchestration frameworks** (LangChain, LangGraph), automating triage, assignment, and status tracking while **reducing manual coordination overhead** for internal operations teams
- Built backend services for a node-based workflow editor, enabling teams to configure executable multi-step inference and data-processing pipelines
- Integrated a **4-system stack** (PostgreSQL, MongoDB, Redis, NATS) for asynchronous stateful execution across internal workflow endpoints; containerized services with Docker for reproducible deployments

IIITDM Jabalpur – Materials Science Lab

Nov 2023 – Apr 2024

Research Intern – Machine Learning (On-site)

Jabalpur, Madhya Pradesh

- Benchmarked **5 ML architectures** (GNNs, XGBoost, Random Forest, DNNs) on semiconductor band-gap prediction, achieving **83% accuracy**
- Designed an automated data-collection and preprocessing pipeline ingesting **1,600+ material records** from the Materials Project API across all 118 elements
- Evaluated model performance using MSE, MAE, and R^2 to compare architectures and select robust baselines
- Contributed to research abstract accepted at **Materials Today Conference (Elsevier, 2025)**

PROJECTS

MatterGen Python Module 🐍 📦

Mar 2026

- Developed and **published to PyPI** a Python wrapper for Microsoft's MatterGen diffusion model for materials design
- Implemented APIs for model loading and property-conditioned generation (e.g., band gap, bulk modulus) to produce candidate materials for downstream screening

Thermoelectric Material Property Prediction 🔬

Jan 2025 – May 2025

- Benchmarked **7 ML algorithms** (XGBoost, Random Forest, ResNet, FFNN, ensembles) on **7,239** thermoelectric materials
- Showed chemical descriptors outperform physical ones by **25% in R^2** , yielding a feature-engineering heuristic for screening

NLP Classifiers – Language Detection & Sentence Entailment 🗣️ 🔄

Jul 2024 – Aug 2024

- Fine-tuned BERT and XLM-RoBERTa, achieving **98% accuracy** on multilingual detection across **20+ languages**
- Built a 3-class entailment classifier reaching **78% F1-score**; recovered minority-class recall by **12 points** using weighted loss

PUBLICATIONS & PATENTS

Materials Today Conference (Elsevier, 2025) — Contributed to accepted abstract on ML-based material property prediction
Patent Application Filed — Co-filed **1 patent application** from academic project work (under review)

TECHNICAL SKILLS

Languages: Python, Go, SQL

Frameworks: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph

Core ML and DL: Deep Learning, GNNs, NLP, Transformers, Feature Engineering, Model Evaluation, Cross-Validation

MLOps & Backend: Docker, GCP, REST APIs, Go Fiber, NATS, Git, Linux, CI/CD

Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, MongoDB, Redis

LEADERSHIP

Team Lead, IIITDM Jabalpur — Led teams of 4–6 peers across **3 projects** in Product Realization, Engineering Design, and Cyber-Physical Production Systems; delivered prototypes through design reviews (grades **A**, **B+**, **A+**).